pfi

Python for Fixed Income

COURSE 5C · RISK MODELS

Case Study: Monitoring a Credit Model

WEEK 2 · DAY 4

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Today: a model in use

The reference: study years 1 to 10

10,705
firm-years

217
defaults

Synthetic cohort. Years 11 to 14 are monitored; 15 to 20 stay closed until the capstone.

Four checks a year

Has the score drifted?
Have the inputs?
Does it still rank the defaults first?
Is its count the right size?

Case study: inputs that drift while the ranking holds, and the reverse, planted in a copy.

Names for a check

The model's output is the model's default chance; its sum over a year, the model's count. A red status is a check that asks for a look.

Allowed names

fitted hazard

a synthetic firm's default chance

survival in study years

Never called

a rating view

a real issuer's default chance

a forecast of defaults

Red is a number outside a band written down in advance: never a credit decision.

How monitoring works

The reference	The model fitted on study years 1 to 10, its coefficients frozen
Each later year	Scored by the frozen model, as a desk scores each new year
Input checks	Need only inputs and scores: they can run on the year's first day
Outcome checks	Need the year's defaults: they run after the year is over
What comes out	One row per year, then one report per year

The reference count matches

217

defaults counted, years 1 to 10

217.0000

the model's count

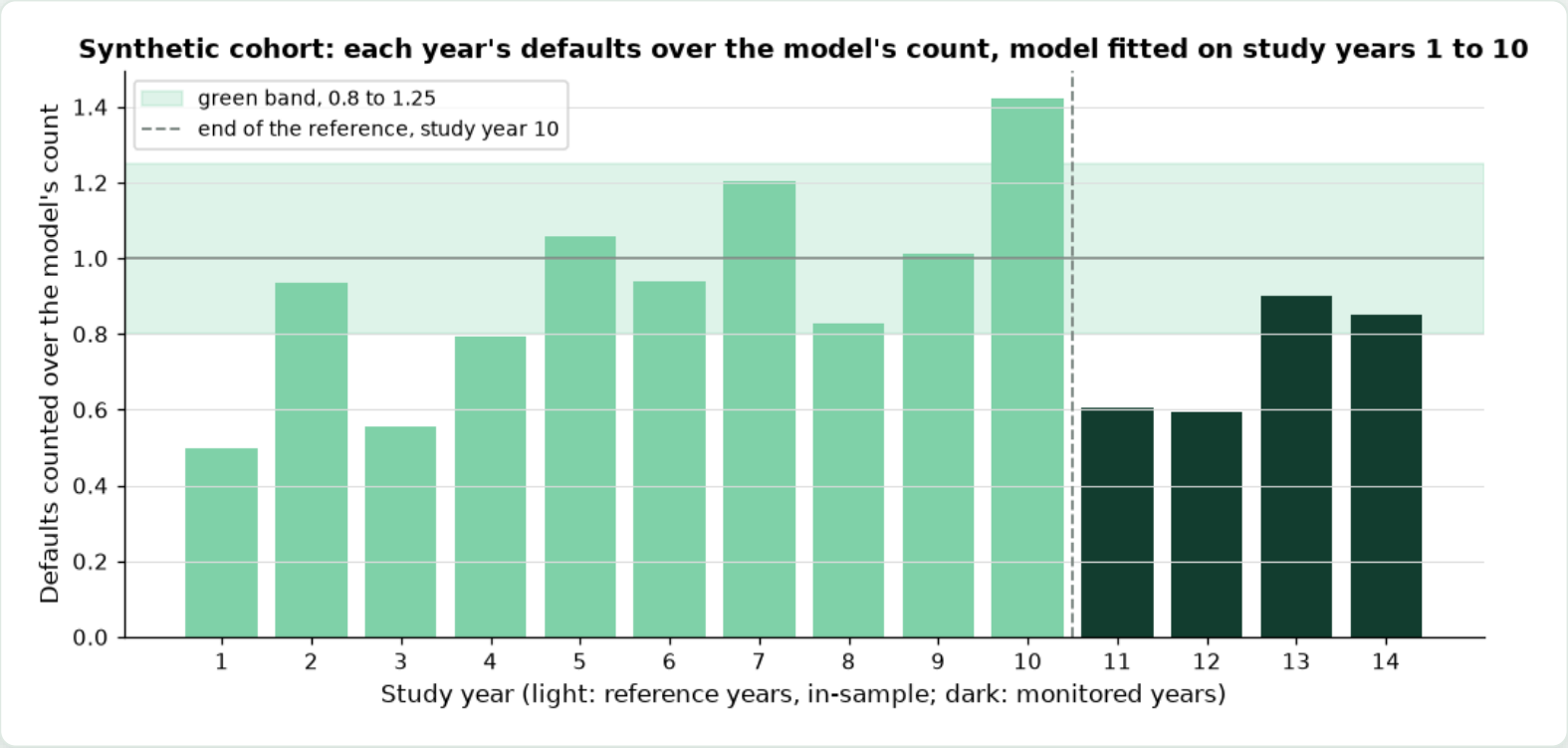
0.869

AUC, in-sample

A logistic regression with a constant makes the sum of its fitted chances equal the count of events on the rows it was fitted on. Only later years can show a gap.

In-sample numbers are a yardstick, not evidence the model works.

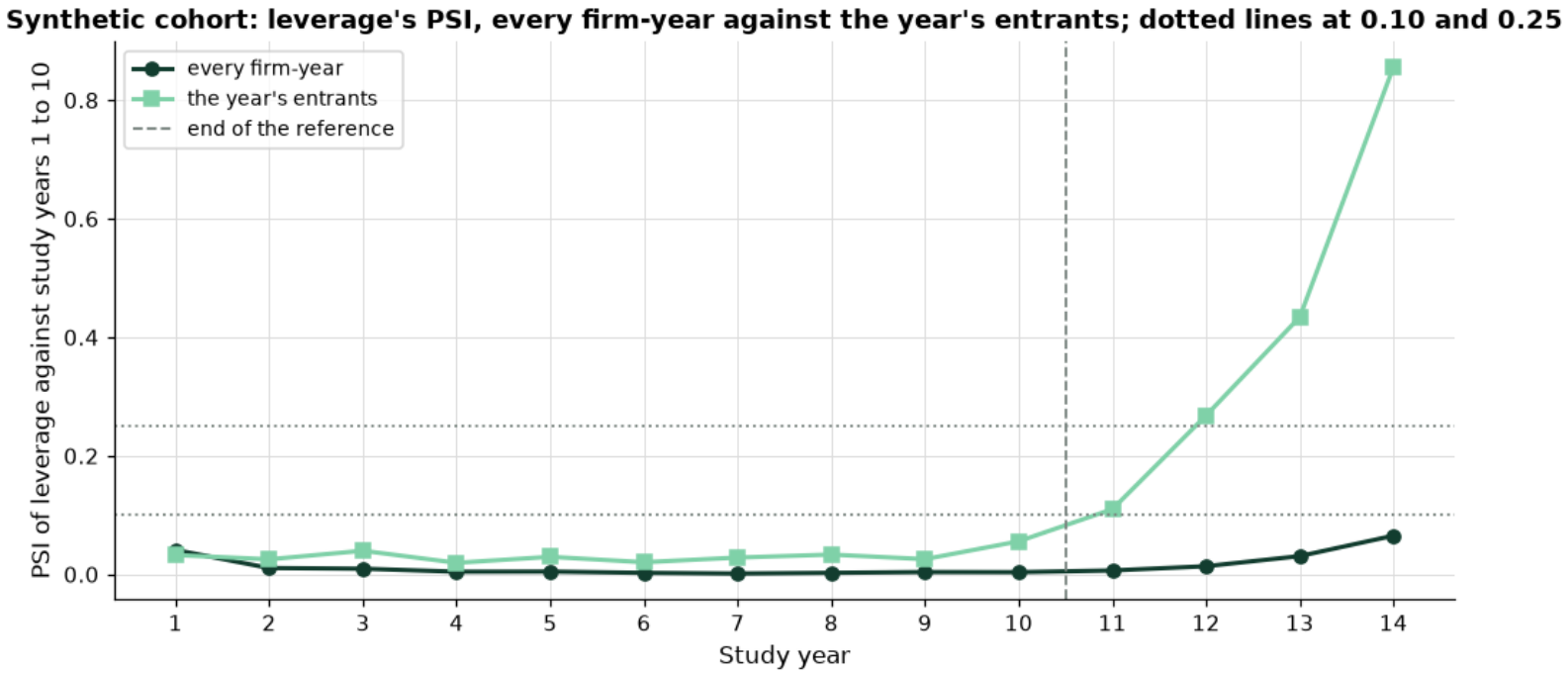
Counted over fitted



Year	Ratio	Chance	AUC
11	0.604	0.010	0.894
12	0.592	0.004	0.917
13	0.902	0.288	0.883
14	0.851	0.155	0.877

Chance: of so few defaults or fewer, if the model's count were the true average (Poisson).

Drift: the stock and the flow



Every PSI of the whole book stays green. The new firms drift fast, but they are about one firm-year in eight, so the book follows slowly.

How monitoring_report works

PSI rows	The score and each input; green below 0.10, amber below 0.25, red from 0.25
AUC	Green if it fell by less than 0.03 from the reference, amber by less than 0.07
Count	Counted over fitted: green 0.8 to 1.25, amber 0.67 to 1.5, red outside
The bands	The course's choice for teaching, written in the docstring, never tuned on data
What comes out	Reference, current, and status for every check (monitoring_report)

One report per year

Check	Year 11	Year 12	Year 13	Year 14
PSI: score	green	green	green	green
PSI: rating_score	green	green	green	green
PSI: leverage	green	green	green	green
PSI: log_coverage	green	green	green	green
AUC	green	green	green	green
counted over fitted	red	red	green	green

The report finds the years. It has no column that could say why: the year factor is in no input.

Case study: two cases

Case A: the new firms

Only the firms that entered in study years 11 to 14, whose inputs drifted on day 8. Does the ranking hold?

Case B: planted in a copy

Each year's rating scores shuffled among its firm-years: every input keeps its distribution, the link to default is cut.

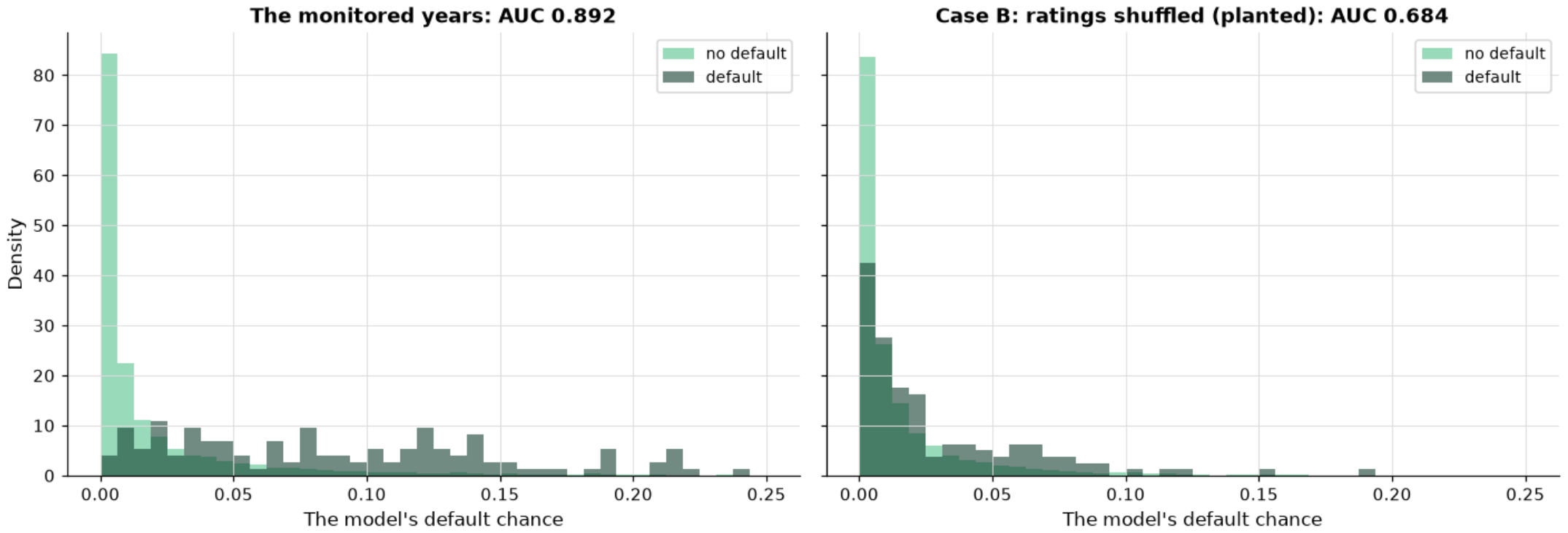
Case B shows what a broken link looks like, not that one happened.

Three reports side by side

Check	Whole book	A: new firms	B: shuffled (planted)
PSI: score	0.005 green	0.245 amber	0.073 green
PSI: rating_score	0.009 green	0.210 amber	0.009 green
PSI: leverage	0.023 green	0.260 red	0.023 green
PSI: log_coverage	0.005 green	0.087 green	0.005 green
AUC	0.892 green	0.880 green	0.684 red
counted over fitted	0.757 amber	0.772 amber	1.065 green

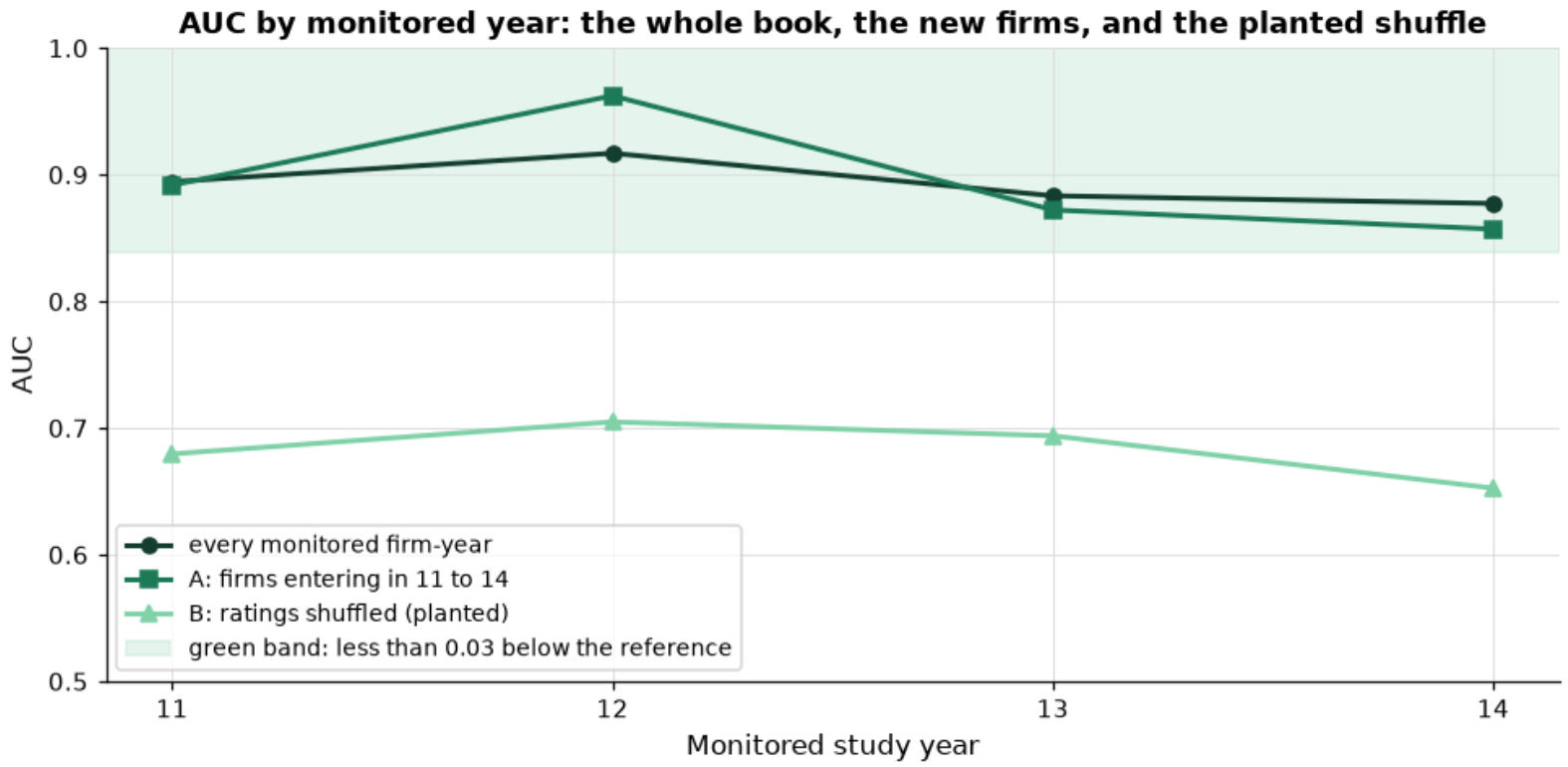
A drifting input is a reason to look, not proof of failure. A stable input is no proof of health.

A broken ranking



No input check can see this. Only the outcome can, and it arrives after the year is over.

AUC by year



The whole book and the new firms stay in the green band every year. The planted copy falls far below it, while its inputs stay green.

What the model shows, and where it fails

What the model shows

Frozen on years 1 to 10, it ranked every monitored year well, and kept ranking the new firms whose inputs drifted most.

On the level	More defaults counted than happened every year; red in 11 and 12
On what monitoring sees	Input checks miss a broken link: case B's inputs were all green
On timing	The new firms' drift reached the whole book slowly

The guidance changed this year

Federal Reserve, OCC, and FDIC, April 17, 2026

SR 26-2: Revised Guidance on Model Risk Management

It "supersedes and replaces SR letter 11-7" (April 4, 2011)

"Ongoing model monitoring involves an evaluation of the extent to which a model is performing as expected"; outcomes analysis "compares model outputs to corresponding real-world outcomes".

Source: federalreserve.gov/supervisionreg/srletters/sr2602.htm and its attachment, section V (read 2026-10-24).

Excel side by side

Excel side by side (tested in Excel)

`=SUM(B2:B1793)/SUM(A2:A1793)`	0.6039
`=POISSON.DIST(SUM(B2:B1793),SUM(A2:A1793),TRUE)`	0.0099

The AUC from ranks: 0.8944

```
=(SUMPRODUCT(RANK.AVG(A2:A1793,A2:A1793,1),B2:B1793)-SUM(B2:B1793)*(SUM(B2:B1793)+1)/2)/(SUM(B2:B1793)*(ROWS(B2:B1793)-SUM(B2:B1793)))
```

No AUC function: the defaults' rank sum, less n(n + 1)/2, over the pairs. Every value matched Python.

Save the day: tag the model

```
git tag monitored-model-v1
git log --oneline --decorate
```

A tag names the commit of the frozen model. Unlike a branch, it never moves.

Which model were the red years checked against? The tag answers.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

What is drawn, from which inputs, in which unit: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: a model fitted on the years it monitors: scores moved by up to 1.90e-01 when only those years' defaults changed.

Code that runs is not the same as code that is right.

What you can do now

- Freeze a model; its in-sample count matches exactly
- Score each year: AUC, counted over fitted, its chance
- Drift of the score and inputs; the stock and the flow
- monitoring_report with bands you state
- Drift without failure, and failure without drift; 35 tests

Week 2, Day 5 The capstone: the held-out years, opened once